

AIRPORT PROPERTY MANAGEMENT SYSTEM

ReadMe Document

Tech Stack & Architecture

Runtime Environment	: Node.js
Framework	: React
Datalabase Engine	: MySQL
Authentication	: JSON Web Tokens(JWT)]

System Architecture & API Endpoints

The backend handles the core routing and relational database communication injectively mapping the following route modules.

1. Authentication Router (/api/auth)

- **POST/api/auth/login** - Authentication administrative user credentials and returns a secure JWT payload valid for 1 hour.

2. Corporate Ledger Router(/api/companies)

- **GET/api/auth/login** - Fetches all registered corporations to display inside the ledger table
- **POST/api/companies** - Creates a brand new corporation profile record. Accepts 'companyName', 'corporateEmail', 'headquartersAddress', and 'contactHotline'.
- **PUT/api/companies/:id** - Modifies existing corporate details or toggles status between 'Active' and 'Inactive'.
- **DELETE/api/companies/:id** - Permanently clears a corporate entry from the database.

3. Shop Profiling Router(/api/shops)

- **GET/api/shops** - Pulls the master shop list showing active sub-shops linked alongside their corporate owners.
- **POST/api/shops** - Profiles and saves a new sub-shop layout registry row. Accepts system parameters like integration types and doc counts.

4. Property Infrastructure Router(/api/properties)

- **GET/api/properties** - Pulls the baseline data rows representing terminal properties and structural slots.
- **POST/api/properties** - Adds a new vacant physical room or zone configuration layer.
- **PUT/api/properties/:id** - Performs dynamic inline row modifications (e.g., updating 'leasedStatus' from No (Available) to Yes (Leased)).

5. Agreement Engine Router(/api/agreement)

- **GET/api/agreements** - Tracks active contract durations, monthly baseline statements, and deposit expirations.
- **POST/api/agreements** - Registers a new binding tenancy contract framework. Captures: 'minimumStoLevel', 'bankDeposit', 'monthlyAmount' (Fixed Rent), and 'confeeRate' (Concession Fee/Commission %).

6. Transaction & Audit Routers(/api/sales)

- **app.use('/api/sales', salesRoutes)** - Captures and writes daily operational cash numbers streaming from the duty-free retail points. Handles 'POST /api/sales' to submit live point-of-sale invoice entries.
- **App.use('/api/sales', reportRoutes)** - Queries database logs using specific dates to generate financial audits and calculations based on the Confee Rate % to show airport authority commission yields.

Local Workspace Installation Guide

1.Core System Prerequisites

Ensure the following software components are activated in your local computer workspace.

1. Node.js (LTS Version): Engine to manage runtime dependencies.
2. XAMPP Control Panel: Local Apache and MySQL server instance ecosystem.

2.Database Creation & Scheme Configuration

1. Spin up the XAMPP Control Panel and execute the Apache and MySQL server layers.
2. Load the phpMyAdmin interface: 'http://localhost/phpmyadmin/'
3. Create a fresh database named precisely: 'dbdutyfreepos'
4. Import your designated structural database backup '.sql' file into this schema.

3.Backend Deployment Execution

Open your terminal window, navigate to the target backend repository, and run these commands.

```
cd backend
npm install
node server.js
```

4.Frontend Deployment Execution

Open your terminal window, navigate to the target frontend repository, and run these commands.

```
cd frontend
npm install
npm run dev
```

